Uttkarsh Ruparel

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Education

Symbiosis Institute of Technology July 2023 - July 2027
Bachelor of Technology in Computer Science (CGPA: 7.68 / 10) *Pune, Maharashtra*

Excellencia Junior College May 2021 - May 2023
Higher Secondary Education / Grade XII (Percentage: 85.5%) *Secunderabad, Telangana*

Pallavi Model School March 2020 - March 2021
Secondary Education / Grade X (Percentage: 85.5%) *Secunderabad, Telangana*

Experience

TraceLink September 2026 – Present
Software Engineer Intern *Pune, Maharashtra*

- Develop and maintain backend services on the **Opus Platform** in the Solution Design (**OSE**) team, supporting multi-enterprise supply chain workflows.
- Build **GraphQL** and **REST** APIs in **Node.js**/**Express**, backed by **Kafka** and **Redis**.
- Ship via **Jenkins** CI/CD as **Docker** containers on **Kubernetes** and **AWS**, and debug with **Jaeger** traces.

Projects

ESX — Stock Exchange Infrastructure | *Go, Kafka, gRPC, PostgreSQL, Redis, AWS (EKS, MSK), Kubernetes*

- Engineered a production-grade securities exchange — **8 microservices** (order gateway, settlement, matching engine, clearing house) over **Kafka** and **gRPC**, replicating core systems behind **NASDAQ**.
- Built a high-concurrency order gateway in **Go** sustaining **6,000+ RPS** at **sub-40ms latency** via **gRPC streaming** and a **Buffer & Drain pattern** (**500k-capacity buffered channels**) to decouple ingestion from matching, surviving **300% load spikes**.
- Implemented atomic **DvP settlement** with an **idempotent double-entry ledger**, preventing duplicate transactions via unique ID tracking and atomic **PostgreSQL upserts** — all movements net to zero.

Hatch — Self-hosted Cloud Deployment Platform | *Go (Gin), AWS (ECS, Fargate), RabbitMQ, Redis, PostgreSQL, Terraform, Docker*

- Architected a self-hosted Render/Railway alternative on **AWS** — independent **Go** microservices for cloning, building and deploying containers on **ECS Fargate**, communicating through **RabbitMQ** with **real-time log streaming** via **Redis pub/sub** over **WebSockets**.
- Achieved **3,500 RPS** at **p99** $<$ **75ms** with zero failed requests across 5,000 requests (ApacheBench) and sustained **716 req/s** at **p95** $<$ **9ms** under 50 concurrent VUs (**k6**).

Kprobe — Kernel Observability for Financial Systems | *Rust, Aya, eBPF, Go, Kafka, Neo4j, ClickHouse, OpenTelemetry, Jaeger*

- Built a **Rust/Aya eBPF probe** attaching to kernel tracepoints at runtime — capturing **TCP RTT**, **CPU scheduling**, and **syscall latency** at nanosecond resolution with zero application instrumentation.
- Engineered a **Go/Kafka** stream processor that windows events into **100ms frames**, constructs a **directed causal graph** by PID/transaction co-occurrence within **50ms**, and classifies chains (`SCHED_DELAY`, `MEMORY_PRESSURE`, `TCP_CHAIN`) into **Neo4j** for **root cause attribution**.
- Instrumented the causal engine, replay engine, and gRPC API server with **OpenTelemetry**, exporting distributed traces to **Jaeger** to debug latency and cross-service failures within kprobe's own Go service mesh.

Technical Skills

Languages: Go, JavaScript, TypeScript, Java, Python, Rust
Backend & Systems: Gin, gRPC, Protocol Buffers, FIX Protocol, Kafka, RabbitMQ, PostgreSQL, sqlc, MongoDB, Redis, ClickHouse, Neo4j, FastAPI, Node.js (Express, Fastify), NGINX, Kong, WebSockets, REST, GraphQL
Infrastructure & Observability: Docker, Kubernetes, OpenShift, ELK Stack, OpenTelemetry, Jaeger, Prometheus, Grafana, Terraform, Ansible, Helm, ArgoCD, CI/CD (GitHub Actions, Jenkins), AWS (MSK, ECR, SQS), k6, Linux, Cloudflare
Frontend: React, Next.js, Astro, Vite, Tailwind CSS
Foundations: System Design, Distributed Systems, Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Database Management Systems